Numerical Methods: The Spectral Eigenvalue Solver

Assembly and solution of the compressible boundary-layer stability eigenvalue problem by Chebyshev collocation in `pyMack`

`pyMack` technical documentation

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Contents

1 Scope: the problem to be solved and the two routes

The companion disturbance-equations document closes with three objects: the four linearised disturbance ODEs for $\hat{q}(y) = (\hat{u}, \hat{v}, \hat{T}, \hat{p})$, their operator form $\mathsf{L}(\alpha; y, \mathsf{D})\,\hat{q} = c\,\mathsf{M}(\alpha; y)\,\hat{q}$, and, built from the Chebyshev differentiation matrices D_1, D_2 constructed there, its discrete image: the finite generalised eigenvalue problem $A(\alpha)\boldsymbol{\phi} = c\,B(\alpha)\,\boldsymbol{\phi}$ on the wall-clustered collocation grid. The present document supplies everything between that statement and a trusted growth rate: the block-by-block assembly, the boundary-condition rows, the eigenvalue algorithms for the temporal and spatial closures, the identification of the physical mode within the computed spectrum, and the sweeps that turn eigenvalues into neutral curves and N-factors.

What is sought. For fixed flow parameters (M_e, Re) the disturbance equations admit non-trivial solutions only at isolated eigenvalues, and these eigenvalues are the entire objective of the computation:

- *Temporal problem*: given a real wavenumber α, find the complex phase speed c and mode shape; the temporal growth rate is $\omega_i = \alpha c_i$.

- *Spatial problem*: given a real frequency ω, find the complex wavenumber α and mode shape; the spatial growth rate is $\sigma = -\alpha_i$.

Every quantity the framework ultimately delivers is assembled from repeated eigenvalue solves: the *neutral curve* is the zero-growth locus $c_i(R, \alpha) = 0$ traced over a parameter sweep; the *N-factor* is the streamwise integral of the spatial growth rate at fixed physical frequency; and the transition estimate is the empirical correlation $N \approx 9$–11 [11, 12]. The numerical task is therefore to evaluate eigenpairs reliably, including the rejection of spurious eigenvalues, and cheaply enough to sweep them by the thousands.

Two routes. Numerical methods for this problem divide into two classical families [2]: *boundary-value methods*, which discretise the ODEs globally and obtain all eigenvalues of the resulting matrix pencil at once without prior knowledge of the spectrum, and *initial-value (shooting) methods*, which integrate the ODEs across the layer and locate one eigenvalue by enforcing the boundary conditions, requiring a good initial guess but very little memory. `pyMack` implements one of each. The present document is the boundary-value route: Chebyshev collocation, block assembly, and dense eigensolves, used for spectrum surveys, neutral curves, and N-factors. Every equation, coefficient, and default below is stated exactly as implemented.

Notation. Overbars denote mean (base-flow) quantities; hats denote complex disturbance amplitudes. The continuous amplitude is $\hat{q}(y) = (\hat{u}, \hat{v}, \hat{T}, \hat{p})$; its discrete (stacked, grid-sampled) form is $\boldsymbol{\phi}$. A prime on a mean quantity denotes $\mathrm{D} \equiv \mathrm{d}/\mathrm{d}y$. The collocation order is N; the grid has $n = N+1$ points, so the assembled matrices are $4n \times 4n$, the size written $4(n+1) \times 4(n+1)$ in the companion document's indexing.

Base flow & transport		**Disturbance / eigenvalue**	
$\overline{U}, \overline{T}, \overline{\rho}{=}1/\overline{T}$	mean velocity, temp., density	$\hat{q}$; $\boldsymbol{\phi}$	amplitude (continuous; discrete)
$\overline{\mu}, \overline{\kappa}$	viscosity, conductivity	$\hat{u}, \hat{v}, \hat{T}, \hat{p}$	velocity/temp./pressure amplitudes
$M_e,\ Re,\ Pr,\ \gamma$	Mach, Reynolds, Prandtl, c_p/c_v	$\alpha,\ \omega$	wavenumber, frequency
$L^*,\ R{=}\sqrt{Re_x}$	Mack length, Mack Reynolds number	$c{=}\omega/\alpha$	complex phase speed; c_r, c_i its parts
$\nu_R{=}1/(Re\overline{\rho})$	viscous prefactor	$\omega_i{=}\alpha c_i;\ \sigma{=}-\alpha_i$	temporal; spatial growth rate
$C{=}\overline{\mu}/\overline{T}$	Chapman–Rubesin group	$N;\ n{=}N{+}1$	collocation order; grid size

Operator shorthands: D_1, D_2 differentiation matrices; $\mathcal{C}_\kappa, \mathcal{C}_d, d_R$ conduction/dissipation prefactors; $\mathcal{L}_c$ conduction operator; A, B (temporal) and C_0, C_1, C_2 (spatial) operator blocks.

2 Assembling the operator and its boundary conditions

The companion document closes with the operator pencil

$$\mathsf{L}(\alpha; y, \mathrm{D})\,\hat{q} = c\,\mathsf{M}(\alpha; y)\,\hat{q}, \qquad \mathsf{L} = \mathsf{A}_2(y)\,\mathrm{D}^2 + \mathsf{A}_1(\alpha; y)\,\mathrm{D} + \mathsf{A}_0(\alpha; y), \tag{1}$$

with rows ordered (C, X, Y, E) and columns $(\hat{u}, \hat{v}, \hat{T}, \hat{p})$, together with the objects that discretise it: the Chebyshev–Gauss–Lobatto nodes mapped to $y \in [0, y_{\max}]$ by the wall-clustering transform, node $j{=}0$ at the freestream and node $j{=}N$ at the wall, and the physical differentiation matrices D_1, D_2 built on that grid. Two practical choices complete the grid before assembly can begin.

Domain height and resolution. The truncation height must clear the boundary layer by a wide margin and is therefore set as a multiple of the boundary-layer scale, $y_{\max} = m\,(\delta^*/L^*)$, with $m \approx 8$–12 for the wall-trapped second mode and larger ($m \approx 35$–45) for the more diffuse first mode; the bare values $y_{\max} = 6/12$ are only a fallback. The default resolution is $N = 128$ ($n = 129$). Because the method is spectral, the eigenvalue converges exponentially in N [6, 7]; at the default resolution the worked-example eigenvalue of Section 6 is already converged to many digits.

Discretising (1) is then the purely mechanical application of three substitutions on the collocation grid $\{y_j\}_{j=0}^{N}$,

$$\mathrm{D} \;\mapsto\; D_1, \qquad \mathrm{D}^2 \;\mapsto\; D_2, \qquad f(y) \;\mapsto\; \Lambda(f) = \mathrm{diag}\big(f(y_0), \ldots, f(y_N)\big), \tag{2}$$

together with the stacking of the four sampled amplitudes into the single unknown $\boldsymbol{\phi} = [\hat{u}; \hat{v}; \hat{T}; \hat{p}] \in \mathbb{C}^{4n}$, the discrete eigenvector, which un-stacked is the mode shape sampled on the grid. Applied entrywise, each scalar entry of the 4×4 operator becomes an $n \times n$ block,

$$A_{ij}(\alpha) = \Lambda\big([\mathsf{A}_2]_{ij}\big)\, D_2 + \Lambda\big([\mathsf{A}_1]_{ij}\big)\, D_1 + \Lambda\big([\mathsf{A}_0]_{ij}\big), \qquad B_{ij}(\alpha) = \Lambda\big([\mathsf{M}]_{ij}\big), \tag{3}$$

and the pencil becomes the generalised problem $A\boldsymbol{\phi} = cB\boldsymbol{\phi}$ of Section 3. Equation (3) is the entire assembly: A is built from exactly three ingredients (the differentiation matrices, diagonal samples of mean-flow profiles, and the scalars α, M_e, Re), while B, the discrete image of the algebraic operator M, marks where the eigenvalue resides. All that remains is to record the four coefficient matrices explicitly and to impose the boundary conditions.

The four coefficient matrices. Two of the four are recorded in the companion document:

$$\mathsf{A}_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -\nu_R\overline{\mu} & 0 & 0 & 0 \\ 0 & -\frac{4}{3}\nu_R\overline{\mu} & 0 & 0 \\ 0 & 0 & -\mathcal{C}_\kappa & 0 \end{pmatrix}, \qquad \mathsf{M} = \mathrm{i}\alpha \begin{pmatrix} 0 & 0 & -1/\overline{T} & \gamma M_e^2 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{4}$$

The remaining two follow by sorting every term of the four disturbance equations by derivative order, with all terms collected on the left. With the transport shorthands $m_1 = \frac{\mathrm{d}\mu}{\mathrm{d}T}$, $m_2 = \frac{\mathrm{d}^2\mu}{\mathrm{d}T^2}$, $k_1 = \frac{1}{\kappa}\frac{\mathrm{d}\kappa}{\mathrm{d}T}$, $k_2 = \frac{1}{\kappa}\frac{\mathrm{d}^2\kappa}{\mathrm{d}T^2}$ and the prefactors $\nu_R = 1/(Re\overline{\rho})$, $\mathcal{C}_\kappa = \gamma\overline{\mu}/(Pr\;Re\,\overline{\rho})$, $\mathcal{C}_d = \gamma(\gamma-1)M_e^2/(Re\,\overline{\rho})$,

$$\mathsf{A}_1(\alpha; y) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -\nu_R\overline{\mu}' & -\frac{\mathrm{i}\alpha}{3}\nu_R\overline{\mu} & -\nu_R m_1 \overline{U}' & 0 \\ -\frac{\mathrm{i}\alpha}{3}\nu_R\overline{\mu} & -\frac{4}{3}\nu_R\overline{\mu}' & 0 & \overline{\rho}^{-1} \\ -2\mathcal{C}_d\overline{\mu}\overline{U}' & (\gamma-1)\overline{\rho}^{-1} & -2\mathcal{C}_\kappa k_1 \overline{T}' & 0 \end{pmatrix}, \tag{5}$$

$$\mathsf{A}_0(\alpha; y) = \begin{pmatrix} \mathrm{i}\alpha & -\overline{T}'/\overline{T} & -\mathrm{i}\alpha\overline{U}/\overline{T} & \mathrm{i}\alpha\gamma M_e^2\overline{U} \\ \mathrm{i}\alpha\overline{U} + \frac{4}{3}\alpha^2\nu_R\overline{\mu} & \overline{U}' - \mathrm{i}\alpha\nu_R\overline{\mu}' & -\nu_R\big(m_1\overline{U}'' + m_2\overline{U}'\overline{T}'\big) & \mathrm{i}\alpha\overline{\rho}^{-1} \\ \frac{2\mathrm{i}\alpha}{3}\nu_R\overline{\mu}' & \mathrm{i}\alpha\overline{U} + \alpha^2\nu_R\overline{\mu} & -\mathrm{i}\alpha\nu_R m_1\overline{U}' & 0 \\ \mathrm{i}\alpha(\gamma-1)\overline{\rho}^{-1} & \overline{T}' - 2\mathrm{i}\alpha\mathcal{C}_d\overline{\mu}\overline{U}' & \mathrm{i}\alpha\overline{U} + \mathcal{C}_\kappa\big(\alpha^2 - k_1\overline{T}'' - k_2\overline{T}'^2\big) - \mathcal{C}_d m_1 \overline{U}'^2 & 0 \end{pmatrix}. \tag{6}$$

Equations (4)–(6), inserted into the assembly rule (3), are the complete temporal operator. Note that α enters only as $\mathrm{i}\alpha$ and α^2, so the operator is a quadratic polynomial in α; this is exactly the property exploited in Section 4, where regrouping the same physics by powers of α yields the spatial quadratic pencil. Figure 1 shows the resulting block occupancy; Figure 2 the assembled numbers for a small solved case.

Example: the continuity block-row. Row C of the assembly rule (3) reads, term for term,

$$A_{(C,\hat{u})} = \mathrm{i}\alpha I, \qquad A_{(C,\hat{v})} = D_1 - \Lambda\big(\overline{T}'/\overline{T}\big), \qquad A_{(C,\hat{T})} = -\mathrm{i}\alpha\,\Lambda\big(\overline{U}/\overline{T}\big), \qquad A_{(C,\hat{p})} = \mathrm{i}\alpha\gamma M_e^2\,\Lambda(\overline{U}),$$

$$B_{(C,\hat{T})} = -\mathrm{i}\alpha\,\Lambda\big(1/\overline{T}\big), \qquad B_{(C,\hat{p})} = \mathrm{i}\alpha\gamma M_e^2 I$$

This gives six non-zero blocks, four in A and two in B, each a diagonal matrix or a diagonal matrix times D_1. Every other block-row is read off (4)–(6) identically.

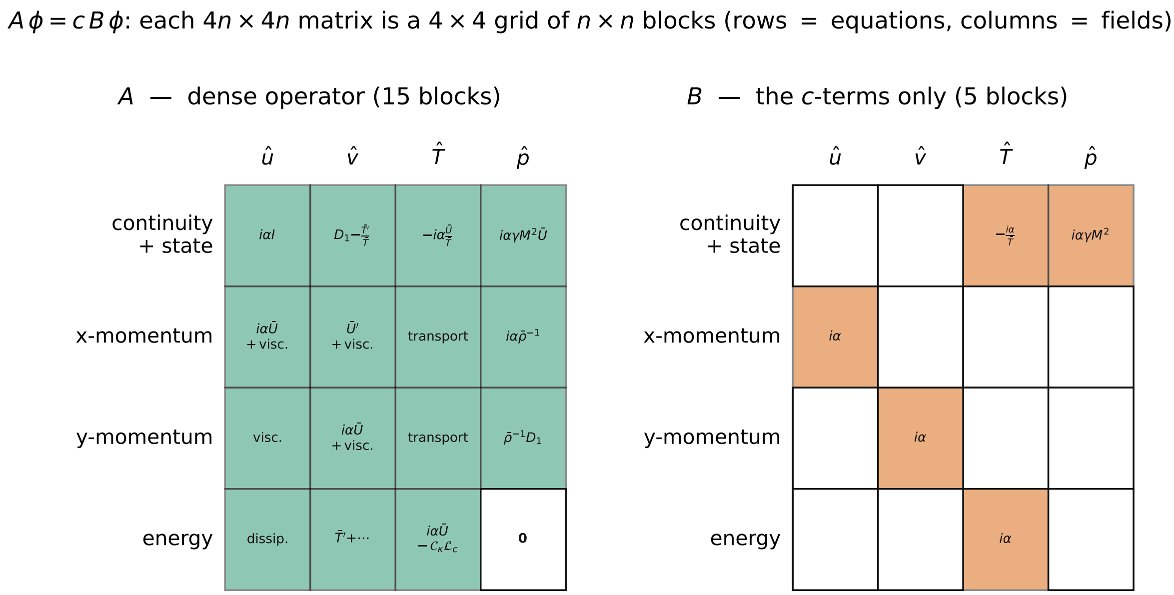

Figure 1: Block structure of $A\boldsymbol{\phi} = cB\boldsymbol{\phi}$. Rows: the four equations (continuity, x-momentum, y-momentum, energy); columns: the four fields $\hat{u}, \hat{v}, \hat{T}, \hat{p}$; shaded = non-zero block. A (left) is block-dense; B (right) holds only five non-zero blocks, the sparsity pattern of M in Eq. (4); every term there carries the eigenvalue c.

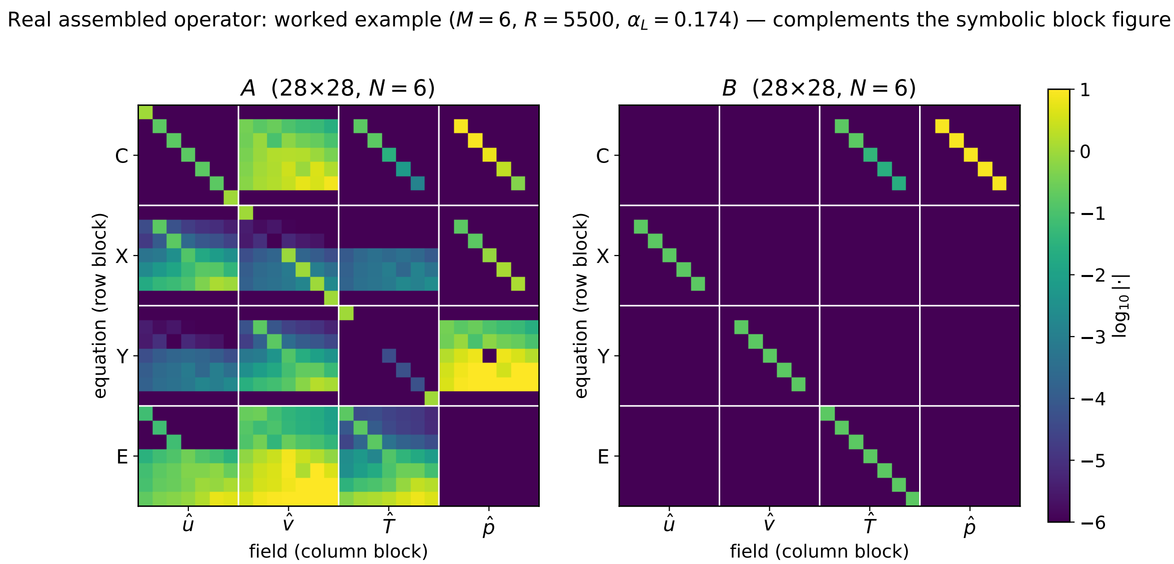

Figure 2: The same block structure assembled numerically ($N = 6$, $4n = 28$, at the parameters of Section 6), shown as $\log_{10}$ of the entry magnitudes. Gridlines every n rows/columns mark the C, X, Y, E row blocks and $\hat{u}, \hat{v}, \hat{T}, \hat{p}$ column blocks of Figure 1: A is densely populated throughout, while B carries only its five algebraic blocks.

Boundary conditions as row replacement. A collocation row states that the equation holds at node y_j; at the wall and freestream those rows are overwritten by the boundary conditions. With the stacking $\boldsymbol{\phi} = [\hat{u}; \hat{v}; \hat{T}; \hat{p}]$, field block $b \in \{0, 1, 2, 3\}$ occupies global indices $bn, \ldots, bn + n - 1$, and, from the Chebyshev node ordering ($\xi_0 = +1$ at $y_{\max}$, $\xi_N = -1$ at the wall), local index 0 is the *freestream* and local index $n-1$ the *wall*:

condition	field	global row	row becomes
no-slip (wall)	$\hat{u}$	$n - 1$	identity ($\hat{u} = 0$)
no-slip (wall)	$\hat{v}$	$2n - 1$	identity ($\hat{v} = 0$)
thermal (wall)	$\hat{T}$	$3n - 1$	identity ($\hat{T} = 0$, isothermal) *or* $D_1[\text{wall}, :]$ ($\mathrm{D}\hat{T} = 0$, adiabatic)
decay (freestream)	$\hat{u}$	0	identity
decay (freestream)	$\hat{v}$	n	identity
decay (freestream)	$\hat{T}$	$2n$	identity

The same rows of B are zeroed, so each constraint is purely algebraic; these zero rows are what make B singular and the problem irreducibly generalised. The pressure $\hat{p}$ carries no explicit boundary row. The inverted node ordering is a genuine implementation hazard: because $\xi_0 = +1 \rightarrow y_{\max}$ and $\xi_N = -1 \rightarrow y = 0$, index 0 is the freestream and $n-1$ the wall, the opposite of the naive guess, and exchanging them produces plausible-looking but wrong eigenvalues (Figure 3).

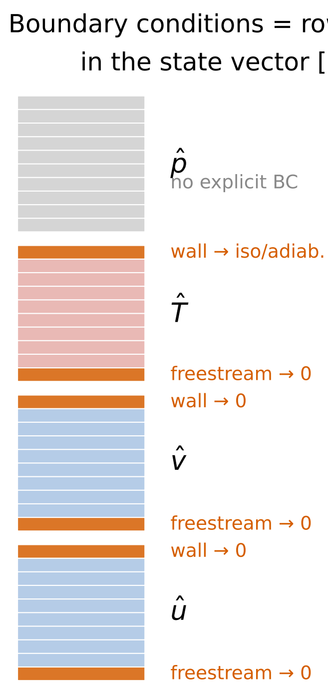

Figure 3: Boundary conditions as row replacement: the stacked state $[\hat{u}; \hat{v}; \hat{T}; \hat{p}]$ with the six overwritten rows highlighted (wall and freestream for $\hat{u}, \hat{v}, \hat{T}$); $\hat{p}$ carries no boundary row. Node index 0 is the freestream and $n-1$ the wall.

3 The temporal problem: a generalised eigenvalue problem

Given a real wavenumber α, the task is to find the complex phase speed c and mode shape $\boldsymbol{\phi}$; the growth rate follows as $\omega_i = \alpha c_i$. With A, B from Section 2 this is the generalised eigenvalue problem

$$A\boldsymbol{\phi} = cB\boldsymbol{\phi}, \qquad A, B \in \mathbb{C}^{4n\times 4n}, \tag{7}$$

solved directly by the QZ algorithm [8], which returns all $4n$ eigenvalues at once; no initial guess is required, the defining advantage of the boundary-value route [2]. The discrete eigenvector $\boldsymbol{\phi}$ is the eigenfunction $\hat{q}(y)$ sampled at the grid points.

A disturbance $\propto e^{\mathrm{i}\alpha(x-ct)}$ with real α behaves as $e^{\alpha c_i t}$ in time, so

$$c_i = \mathrm{Im}(c) > 0 \;\Rightarrow\; \text{unstable (grows in time)}, \qquad c_r = \mathrm{Re}(c) = \text{phase speed}, \tag{8}$$

and the phase speed classifies the mode: a first (Tollmien–Schlichting) mode has $c_r \approx 0.3$–0.6, a second (Mack) mode $c_r \approx 1 - 1/M_e$ [1]. The raw spectrum is filtered to $\mathrm{Re}(c) \in (-0.5, 1.5)$, $|\,\mathrm{Im}(c)| < 0.5$ and sorted by c_i; the identification of the physical mode within the filtered set is the subject of Section 5.

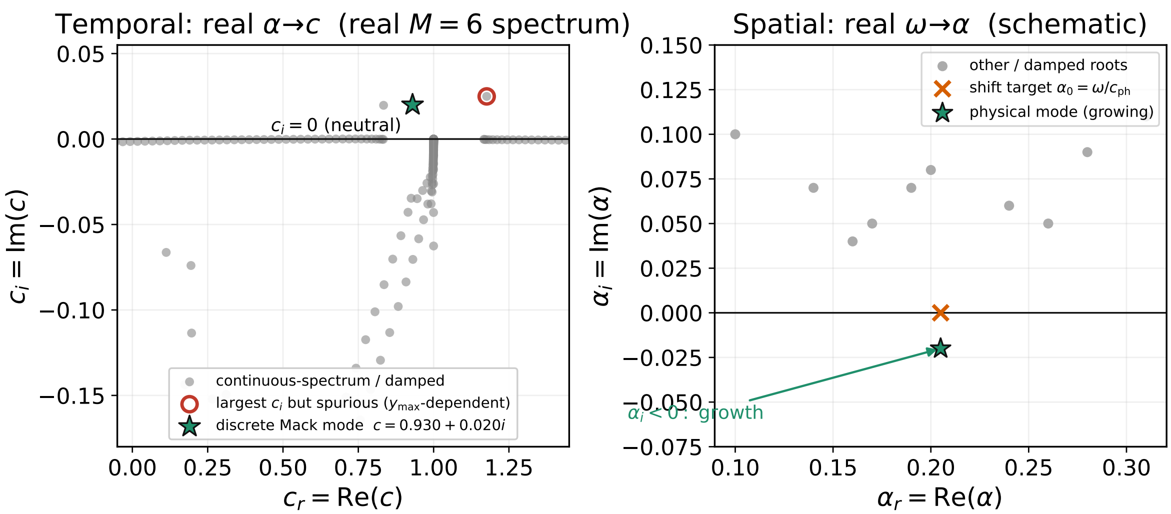

Figure 4: The two eigenvalue planes. *Left (temporal):* input real α; the phase speeds c populate the complex plane; the physical discrete mode sits above the line $c_i = 0$ (unstable) while the continuous spectrum is damped. *Right (spatial, Section 4):* input real ω; the wavenumbers α are sought near a shift target α_0; the physical mode has $\alpha_i < 0$, i.e. spatial growth.

4 The spatial problem: a quadratic eigenvalue problem

Given a real frequency ω (the physically forced case, since a fixed-frequency source generates a disturbance that grows in space), the task is to find the complex wavenumber α; the growth rate follows as $\sigma = -\alpha_i$. Because the viscous and conduction terms carry $\partial_x^2 \mapsto -\alpha^2$, the eigenvalue now appears quadratically. Grouping the discretised equations by powers of α gives the quadratic eigenvalue problem

$$\big(C_0 + \alpha C_1 + \alpha^2 C_2\big)\boldsymbol{\phi} = 0. \tag{9}$$

The spatial closure. The spatial operator is not obtained by re-grouping the temporal listing. It uses the *enthalpy* form of the energy equation, in which $\hat{p}$ appears explicitly through $\mathrm{D}p'/\mathrm{D}t$, and a bulk-viscosity ratio $\lambda/\mu = 1.2$ in place of the Stokes hypothesis, giving the viscous coefficients of Mack [1, Eq. 8.5]

$$c_{\alpha^2} = \tfrac{2(2+\lambda/\mu)}{3} = 2.13, \qquad c_\times = \tfrac{1+2\lambda/\mu}{3} = 1.13, \qquad c_a = \tfrac{2(\lambda/\mu-1)}{3} = 0.13. \tag{10}$$

The two closures differ only in the bulk-viscosity treatment and the energy-equation arrangement; both approximate the same physical mode and agree on it to within discretisation error.

The three coefficient matrices. Because α enters the disturbance equations only as $\mathrm{i}\alpha$ (one streamwise derivative) and α^2 (two), every term of the discretised system belongs to exactly one power of α, and sorting by that power is the entire construction of (9). The three matrices are displayed in the same 4×4 block form as the temporal operator, rows (C, X, Y, E) and columns $(\hat{u}, \hat{v}, \hat{T}, \hat{p})$; entries now contain D_1, D_2 directly, every barred coefficient stands for its diagonal sample $\Lambda(\cdot)$, and I is the identity. Besides m_1, m_2, k_1, k_2 and the prefactors $\nu_R = 1/(Re\,\overline{\rho})$, $d_R = (\gamma-1)M_e^2\nu_R$, two operator shorthands keep the display compact: the viscous and conductive diffusion operators

$$\mathcal{V} = \nu_R\big(\overline{\mu}D_2 + \overline{\mu}'D_1\big), \qquad \mathcal{K} = \nu_R\overline{\kappa}\big(D_2 + 2k_1\overline{T}'D_1 + k_1\overline{T}'' + k_2(\overline{T}')^2\big), \tag{11}$$

where the operator in $\mathcal{K}$ is the conduction operator $\mathcal{L}_c$ of the companion document with its $-\alpha^2$ term removed: that term now lives in C_2.

C_2 collects the $\partial_x^2 \mapsto -\alpha^2$ terms, the streamwise second derivatives of viscous diffusion and heat conduction. It is purely algebraic and has the same occupancy as A_2, with the pressure column empty because there is no pressure diffusion:

$$C_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ c_{\alpha^2}\nu_R\overline{\mu} & 0 & 0 & 0 \\ 0 & \nu_R\overline{\mu} & 0 & 0 \\ 0 & 0 & \nu_R\overline{\kappa} & 0 \end{pmatrix}. \tag{12}$$

C_1 collects every term with a single streamwise derivative: streamwise convection at speed $\overline{U}$, the streamwise pressure gradient, and the mixed $\partial_x\partial_y$ viscous couplings. A common factor i comes out in front:

$$C_1 = \mathrm{i}\begin{pmatrix} I & 0 & -\overline{U}/\overline{T} & \gamma M_e^2\overline{U} \\ \overline{U} & -\nu_R(c_\times\overline{\mu}D_1 + \overline{\mu}') & 0 & \overline{\rho}^{-1} \\ -\nu_R(c_\times\overline{\mu}D_1 + c_a\overline{\mu}') & \overline{U} & -\nu_R m_1\overline{U}' & 0 \\ 0 & -2d_R\overline{\mu}\overline{U}' & \overline{U} & -(\gamma-1)M_e^2\overline{U}\overline{\rho}^{-1} \end{pmatrix}. \tag{13}$$

C_0 carries everything α-free: the wall-normal physics (the D_1, D_2 terms) and the frequency. The eigenvalue-marking role of the temporal B is now played by the iω entries, which sit exactly where the algebraic operator M was non-zero, because $-\mathrm{i}\omega$ is the ansatz image of the time derivative:

$$C_0 = \begin{pmatrix} 0 & D_1 - \overline{T}'/\overline{T} & \mathrm{i}\omega/\overline{T} & -\mathrm{i}\omega\gamma M_e^2 \\ -\mathrm{i}\omega - \mathcal{V} & \overline{U}' & -\nu_R\big(m_1(\overline{U}'' + \overline{U}'D_1) + m_2\overline{U}'\overline{T}'\big) & 0 \\ 0 & -\mathrm{i}\omega - c_{\alpha^2}\mathcal{V} & 0 & \overline{\rho}^{-1}D_1 \\ -2d_R\overline{\mu}\overline{U}'D_1 & \overline{T}' & -\mathrm{i}\omega - \mathcal{K} - d_R m_1(\overline{U}')^2 & \mathrm{i}\omega(\gamma-1)M_e^2\overline{\rho}^{-1} \end{pmatrix}. \tag{14}$$

Equations (12)–(14) are the complete spatial operator: given ω, the mean profiles, and D_1, D_2, the three matrices are assembled block by block exactly as in (3), the boundary rows are replaced as in Section 2, and (9) is solved for the complex wavenumber.

Companion linearisation. A quadratic pencil becomes an ordinary generalised problem of twice the size by introducing the auxiliary variable $\boldsymbol{\psi} = \alpha\boldsymbol{\phi}$ [9]:

$$\underbrace{\begin{bmatrix} -C_1 & -C_0 \\ I & 0 \end{bmatrix}}_{\mathcal{L}} \begin{bmatrix} \boldsymbol{\psi} \\ \boldsymbol{\phi} \end{bmatrix} = \alpha \underbrace{\begin{bmatrix} C_2 & 0 \\ 0 & I \end{bmatrix}}_{\mathcal{R}} \begin{bmatrix} \boldsymbol{\psi} \\ \boldsymbol{\phi} \end{bmatrix}. \tag{15}$$

The lower block row enforces $\boldsymbol{\psi} = \alpha\boldsymbol{\phi}$; the upper row then reproduces (9). The eigenvalue is α and the physical shape is the lower half $\boldsymbol{\phi}$: since $\boldsymbol{\psi} = \alpha\boldsymbol{\phi}$ the two halves are parallel, and by convention the lower half is normalised as the eigenfunction (Figure 5).

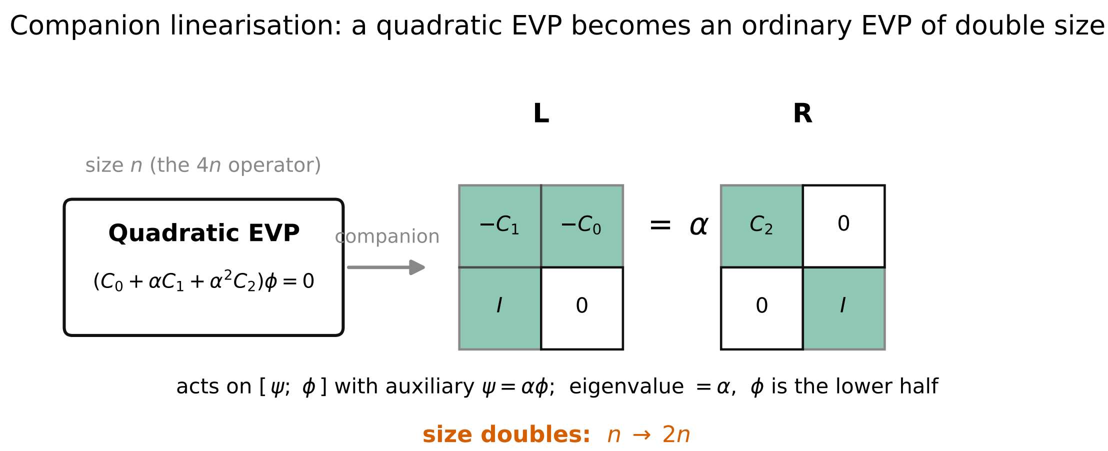

Figure 5: Companion linearisation (15): the $4n$-sized quadratic problem becomes an ordinary generalised problem of size $8n$ in $[\boldsymbol{\psi}; \boldsymbol{\phi}]$ with $\boldsymbol{\psi} = \alpha\boldsymbol{\phi}$. Its eigenvalue is exactly α; the eigenfunction is the lower half.

Shift–invert. The physically relevant branch is targeted with a shift $\alpha_0 = \omega/c_{\text{ph}}$ ($c_{\text{ph}} = 1 - 1/M_e$ for $M_e > 3$, else ≈ 0.4; these are the second- and first-mode phase-speed estimates of Section 3). Subtracting α_0 and inverting maps eigenvalues near α_0 to large μ:

$$\text{eigenvalues of } (\mathcal{L} - \alpha_0\mathcal{R})^{-1}\mathcal{R} \;\text{ are }\; \mu = \frac{1}{\alpha - \alpha_0}, \qquad \Rightarrow \qquad \alpha = \alpha_0 + \frac{1}{\mu}, \tag{16}$$

so the largest-$|\mu|$ modes are the wavenumbers nearest the physical one (in `pyMack`, a dense solve followed by selecting the 20 eigenvalues nearest α_0 and sorting by $\operatorname{Im}\alpha$). The amplitude varies as $e^{\mathrm{i}\alpha x} = e^{\mathrm{i}\alpha_r x}e^{-\alpha_i x}$, so

$$\sigma = -\operatorname{Im}(\alpha) = -\alpha_i > 0 \;\Rightarrow\; \text{unstable (grows downstream).} \tag{17}$$

Alternative route: Newton iteration from a temporal seed. When shift–invert struggles, a more robust path solves the *temporal* problem at a real trial wavenumber $\alpha_r = \omega/c_{\text{est}}$ ($c_{\text{est}} = 0.98$ for $M_e > 3$, else 0.35), keeping only modes that agree across two resolutions (tolerance 8×10^{-3}); forms the seed $\operatorname{Im}(\alpha) \approx -\alpha_r c_i/c_r$, a phase-speed estimate of the rigorous group-velocity conversion of Gaster [10] that suffices here because it only seeds the iteration, and then Newton-iterates the dispersion residual $F(\alpha) = \alpha c(\alpha) - \omega = 0$, $\alpha^{(k+1)} = \alpha^{(k)} - F/F'$, with F' approximated by a finite difference (step 10^{-5}, tolerance 10^{-6}, at most 20 iterations). Both routes return the same $\sigma = -\alpha_i$.

5 Identifying the physical mode

A single solve of (7) returns $4n$ eigenvalues, several hundred at the default resolution, of which typically *one* is the boundary-layer mode sought. The remainder fall into three recognisable populations:

- **Infinite eigenvalues.** The boundary-condition rows carry no eigenvalue and leave zero rows in B, so QZ reports the corresponding pairs as infinite or absurdly large; the raw band filter of Section 3 removes them.

- **The discretised continuous spectrum.** On the unbounded domain the freestream solutions form continuous branches rather than isolated points; the finite grid renders those branches as dense strings of grid-dependent eigenvalues.
- **Discrete modes.** A handful of isolated eigenvalues whose eigenfunctions are trapped inside the layer; the unstable ones among them are the physics sought.

No property of the eigenvalue alone separates the last two populations: in the worked example of Section 6 the largest growth rate in the entire filtered spectrum belongs to a continuous-spectrum artifact. The separation of the discrete modes from the continuous spectrum rests on the eigen*function* and the grid [3, 4], through two implemented tests.

Test 1: freestream decay. A genuine boundary-layer eigenfunction decays exponentially above the layer, whereas a continuous-spectrum eigenfunction oscillates undiminished to the top of the box [4]. The implemented measure is the edge ratio of the stacked eigenvector,

$$r_e = \frac{\langle a(y_j) \rangle_{4 \text{ outermost nodes}}}{\max_j a(y_j)}, \qquad a(y_j) = \max\left(|\hat{u}_j|, |\hat{v}_j|, |\hat{T}_j|, |\hat{p}_j|\right), \tag{18}$$

and candidates with $r_e \geq 0.1$ are rejected. The test is not marginal. At the worked-example parameters ($M_e = 6$, $R = 5500$, $\alpha_L = 0.174$) the accepted Mack mode has $r_e = 1.7 \times 10^{-7}$, while the two rejected amplified candidates have $r_e = 0.34$ and 0.50: six orders of magnitude of separation.

Test 2: domain stationarity. A genuine eigenvalue is a property of the boundary layer, not of the box that truncates it, so raising $y_{\max}$ must leave it fixed; the continuous-spectrum approximants, which owe their existence to the unbounded domain [3], rearrange with the grid. The solve is repeated on a taller domain and a candidate is kept only if its c reappears there within 2×10^{-3}. At the worked-example parameters, raising $y_{\max}$ from 10 to $12\,(\delta^*/L^*)$ moves the Mack mode by 5×10^{-8} and every rejected candidate by 10^{-3}–10^{-2}.

6 Worked example: one temporal solve

Consider a strong Mack-mode case: $M_e = 6.0$, $R = 5500$, $\alpha_L = 0.174$, adiabatic wall, L^* scaling, $N = 180$ ($n = 181$, matrices $4n = 724$ wide), and domain height $y_{\max} = 10\,(\delta^*/L^*) \approx 128$ (a multiple of the boundary-layer scale, per Section 2). Solving (7) returns 724 eigenvalues; after the physical filter, the six least-stable are listed below with the two acceptance measures of Section 5, the edge ratio r_e and the eigenvalue shift $|\Delta c|$ under $y_{\max} \to 12\,(\delta^*/L^*)$:

c_r	c_i	$\omega_i = \alpha c_i$	r_e	$\lvert\Delta c\rvert$	verdict
+1.176	+0.0249	+0.00434	0.34	6×10^{-3}	fails both: continuous spectrum
+0.930	**+0.0200**	**+0.00348**	$\mathbf{1.7 \times 10^{-7}}$	$\mathbf{5 \times 10^{-8}}$	**discrete Mack mode (the answer)**
+0.834	+0.0197	+0.00343	0.50	5×10^{-3}	fails both: continuous spectrum
+0.740	+0.0000	0.00000	0.072	5×10^{-3}	fails test 2: neutral continuous cluster
+0.720	+0.0000	0.00000	0.075	1.4×10^{-3}	passes both (marginally); neutral, out-grown
+0.760	+0.0000	0.00000	0.070	7×10^{-3}	fails test 2: neutral continuous cluster

This is precisely the trap the tests of Section 5 exist to avoid: the largest-growth eigenvalue (first row, $c_r = 1.18$) is *not* the answer. Its eigenfunction retains a third of its peak amplitude at the top of the box, its c shifts by 6×10^{-3} when the domain is raised, and $c_r > 1$ is unphysically supersonic.

The physical answer is the second row, the Mack (second) mode $c = 0.930 + 0.0200\mathrm{i}$: edge ratio 10^{-7}, domain-stationary to 10^{-7}, with an eigenfunction that decays at the freestream (Figure 6). Its growth rate is $\omega_i = \alpha c_i = +0.00348$ (unstable) and its phase speed $c_r = 0.93$ is the second-mode value near $1 - 1/M_e$ [1]. This single $\omega_i > 0$ is one point inside the neutral curve; repeating the solve across an (R, α) grid and locating the zero crossing of ω_i traces the neutral curve of Section 7.

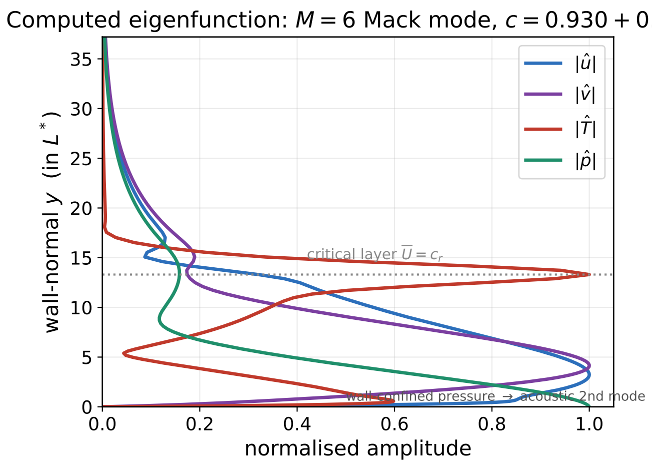

Figure 6: Computed eigenfunction of the worked-example Mack mode: the four amplitude fields $|\hat{u}|, |\hat{v}|, |\hat{T}|, |\hat{p}|$ against y (each normalised). $|\hat{T}|$ peaks at the critical layer ($\overline{U} = c_r$); $|\hat{p}|$ is wall-confined, the acoustic signature of the second mode. All fields decay at the freestream, confirming a discrete mode.

7 Neutral curves

The neutral curve is the boundary between growth and decay: the locus $c_i(R, \alpha) = 0$ (temporal) or $\sigma(R, \omega) = 0$ (spatial), inside which the flow is unstable. `pyMack` traces it in three steps:

1. **Sweep** an (R, α) grid at fixed M_e; at each node solve (7), apply the mode tests of Section 5, and store c_i.
2. **Extract** the zero contour of the stored field by marching squares; unresolved nodes are tagged stable so that the contour closes.
3. **Continuation** for difficult branches: at low Mach number the first-mode eigenvalue approaches the continuous spectrum and the grid sweep loses it. A continuation pass then tracks the mode by proximity: from a clean seed it marches in R, follows the nearest eigenvalue, and secant-solves $c_i = 0$ at each step, each converged eigenvalue seeding the next parameter value.

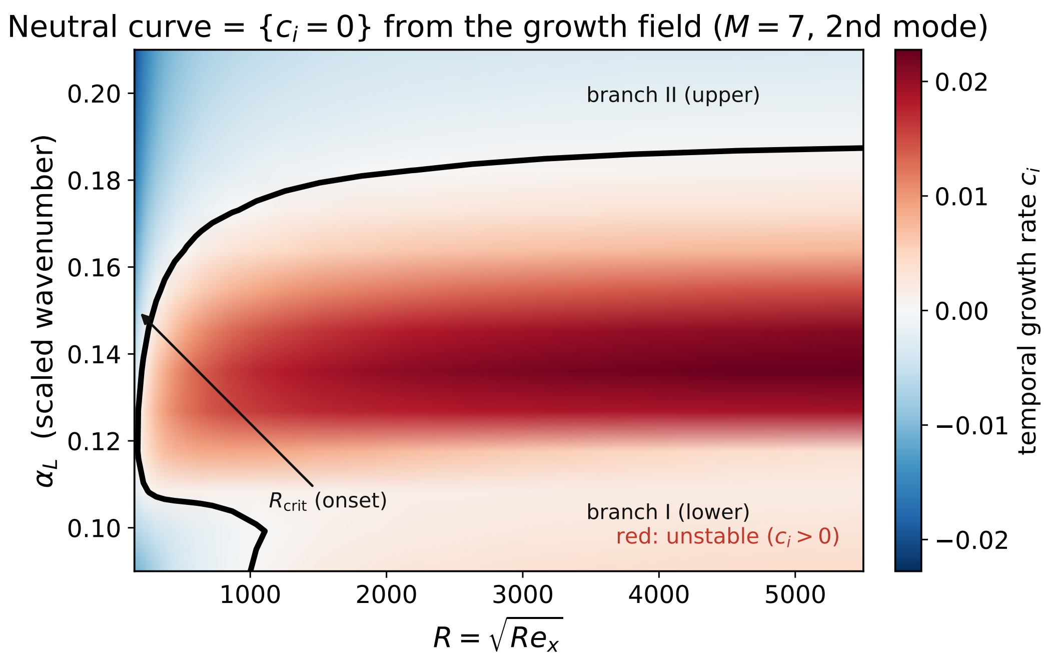

Figure 7: Steps 1–2 of the neutral-curve procedure for a solved case ($M_e = 7$, second mode): the coloured field is the growth rate $c_i(R, \alpha)$ from the sweep; the black curve is the extracted $\{c_i = 0\}$ contour. The flow is unstable inside the loop; R_{crit} marks the lower branch (branch I) where instability first appears as R increases, and the curve closes at branch II.

8 N-factor and transition

What matters for transition is the *accumulated* amplification of a fixed-frequency disturbance. For a chosen reduced frequency F, the spatial problem of Section 4 is solved at successive stations to obtain $\sigma_L = -\operatorname{Im}(\alpha_L)$, the spatial growth rate (17) expressed in the L^* scaling, and integrated from the lower neutral point R_0 where σ_L first becomes positive:

$$N(R) = \int_{R_0}^{R} \max(\sigma_L, 0)\,\mathrm{d}R' \quad \text{(trapezoidal)}, \qquad \frac{A(R)}{A(R_0)} = e^{N(R)}. \tag{19}$$

The amplitude grows by e^N, and transition is empirically expected near $N \approx 9$–11, following the e^N method of Smith & Gamberoni and van Ingen [11, 12]. For a cone, surface arc length is mapped to an equivalent R before integration. The transition N is a correlation, not a first-principles criterion: it depends on the disturbance environment (flight $\sim$9–11; noisy conventional tunnels transition at $N \sim 4$–6), presumes a single dominant linear mode, and absorbs the unknown initial amplitude set by receptivity.

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